Building Diabetes Prediction Models

```
diabetes = readRDS("clean_diabetes_data.rds")
```

Model 1 Preparation

```
df_multilog <- diabetes |>
  select(diabetes_012, high_bp, high_chol, chol_check, bmi, smoker, stroke,
         heart_diseaseor_attack, phys_activity, fruits, veggies, hvy_alcohol_consump,
         any_healthcare, no_docbc_cost, gen_hlth, ment_hlth, phys_hlth, diff_walk,
         sex, age, education, income)
df_multilog
```

```
## # A tibble: 253,680 x 22
##    diabetes_012 high_bp high_chol chol_check   bmi smoker stroke
##    <fct>        <fct>   <fct>     <fct>      <dbl> <fct>  <fct>
##  1 No Diabetes  Yes     Yes       Yes           40 Yes    No
##  2 No Diabetes  No      No        No            25 Yes    No
##  3 No Diabetes  Yes     Yes       Yes           28 No     No
##  4 No Diabetes  Yes     No        Yes           27 No     No
##  5 No Diabetes  Yes     Yes       Yes           24 No     No
##  6 No Diabetes  Yes     Yes       Yes           25 Yes    No
##  7 No Diabetes  Yes     No        Yes           30 Yes    No
##  8 No Diabetes  Yes     Yes       Yes           25 Yes    No
##  9 Diabetes     Yes     Yes       Yes           30 Yes    No
## 10 No Diabetes  No      No        Yes           24 No     No
## # i 253,670 more rows
## # i 15 more variables: heart_diseaseor_attack <fct>, phys_activity <fct>,
## #   fruits <fct>, veggies <fct>, hvy_alcohol_consump <fct>,
## #   any_healthcare <fct>, no_docbc_cost <fct>, gen_hlth <ord>, ment_hlth <dbl>,
## #   phys_hlth <dbl>, diff_walk <fct>, sex <fct>, age <ord>, education <ord>,
## #   income <ord>
```

train/test split

```
# using createDataPartition because diabetes_012 is unbalanced based on EDA.
train_test_split <- createDataPartition(df_multilog$diabetes_012, p = 0.8, list = FALSE)
train <- df_multilog[train_test_split, ]
test  <- df_multilog[-train_test_split, ]

y_test <- test$diabetes_012
```

F1 function for multiclass

```
multi_f1 <- function(true, pred) {
  # Convert to factors
  true <- as.factor(true)
  pred <- as.factor(pred)

  # make sure all classes are taken into consideration (bc diabetes=1 is rare)
  class <- union(levels(true), levels(pred))
  true <- factor(true, levels = class)
  pred <- factor(pred, levels = class)

  # A vector to store F1 for each class
  multi_f1_vec <- numeric(length(class))

  # Loop over each class
  for (j in seq_along(class)) {
    i <- class[j]
    tp <- sum(true == i & pred == i)
    fn <- sum(true == i & pred != i)
    fp <- sum(true != i & pred == i)
    rec  <- if((tp + fn) == 0) 0 else tp / (tp + fn)
    prec <- if((tp + fp) == 0) 0 else tp / (tp + fp)

    if(prec + rec == 0) {
      multi_f1_vec[j] <- 0
    } else {
      multi_f1_vec[j] <- 2 * prec * rec / (prec + rec)
    }
  }
    mean(multi_f1_vec)
}
```

Model 1: Multinomial Logistic Regression

```
model1 <- multinom(diabetes_012 ~ ., data = train, trace = FALSE)
```

```
model1_pred <- predict(model1, newdata = test, type = "class")
model1_prob <- predict(model1, newdata = test, type = "prob")

model1_accuracy <- mean(model1_pred == y_test)
cat("Accuracy is:", model1_accuracy, "\n")
```

```
## Accuracy is: 0.8481719
```

```
model1_macro_f1  <- multi_f1(y_test, model1_pred)
cat("Macro F1 score is:", model1_macro_f1, "\n")
```

```
## Macro F1 score is: 0.39549
```

```r
model1__macro_auc <- as.numeric(multiclass.roc(response = y_test, predictor = model1_prob)$auc)
cat("Macro AUC is:", model1__macro_auc, "\n")
```

```
## Macro AUC is: 0.6955571
```

1.1 ROC

```r
cols <- c("blue", "green", "pink")

# Compute one-vs-rest ROC and AUC for each class
model1_roc1 <- roc(as.numeric(y_test == levels(y_test)[1]), model1_prob[, levels(y_test)[1]])
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```r
model1_roc2 <- roc(as.numeric(y_test == levels(y_test)[2]), model1_prob[, levels(y_test)[2]])
```

```
## Setting levels: control = 0, case = 1
## Setting direction: controls < cases
```

```r
model1_roc3 <- roc(as.numeric(y_test == levels(y_test)[3]), model1_prob[, levels(y_test)[3]])
```

```
## Setting levels: control = 0, case = 1
## Setting direction: controls < cases
```

```r
# Plot ROC
plot(model1_roc1, col = cols[1], main = "ROC - Multinomial Logistic Regression")
lines(model1_roc2, col = cols[2])
lines(model1_roc3, col = cols[3])

# Compute class AUCs
model1_class_aucs <- sapply(levels(y_test), function(cl) round(auc(roc(as.numeric(y_test == cl), model1_
```

```
## Setting levels: control = 0, case = 1
## Setting direction: controls < cases
```

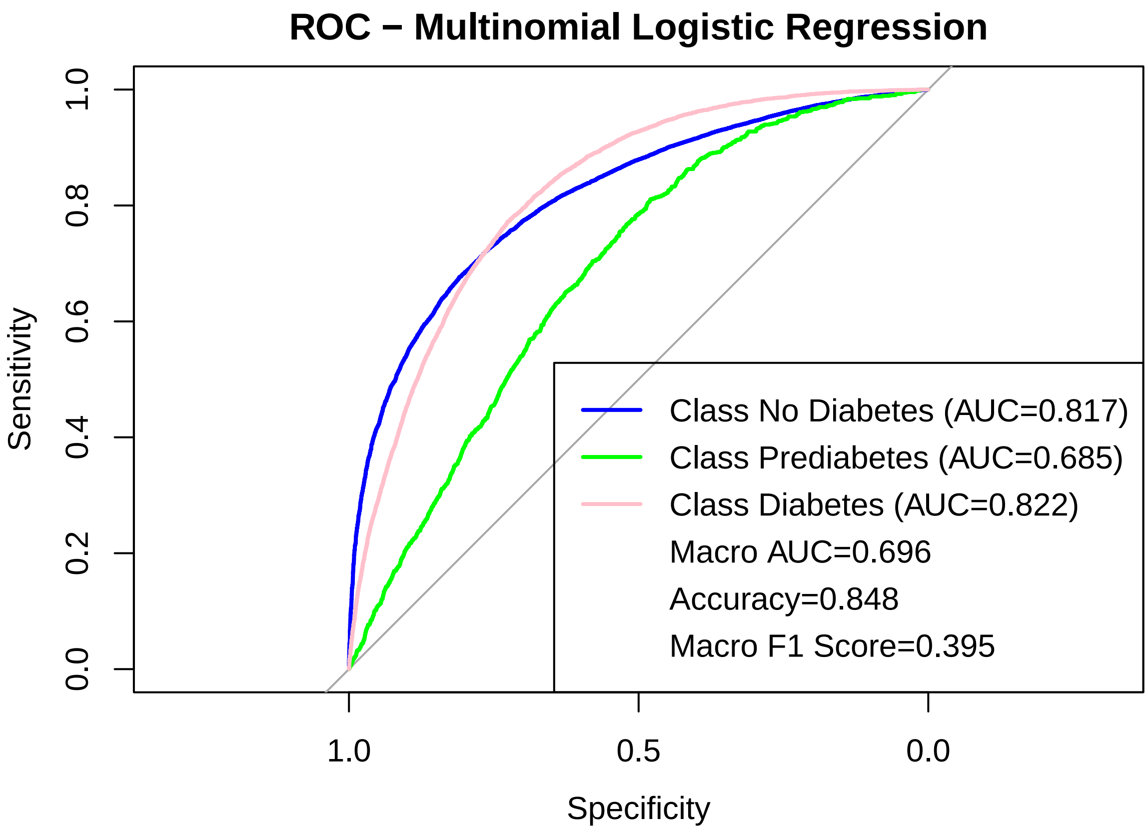
```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

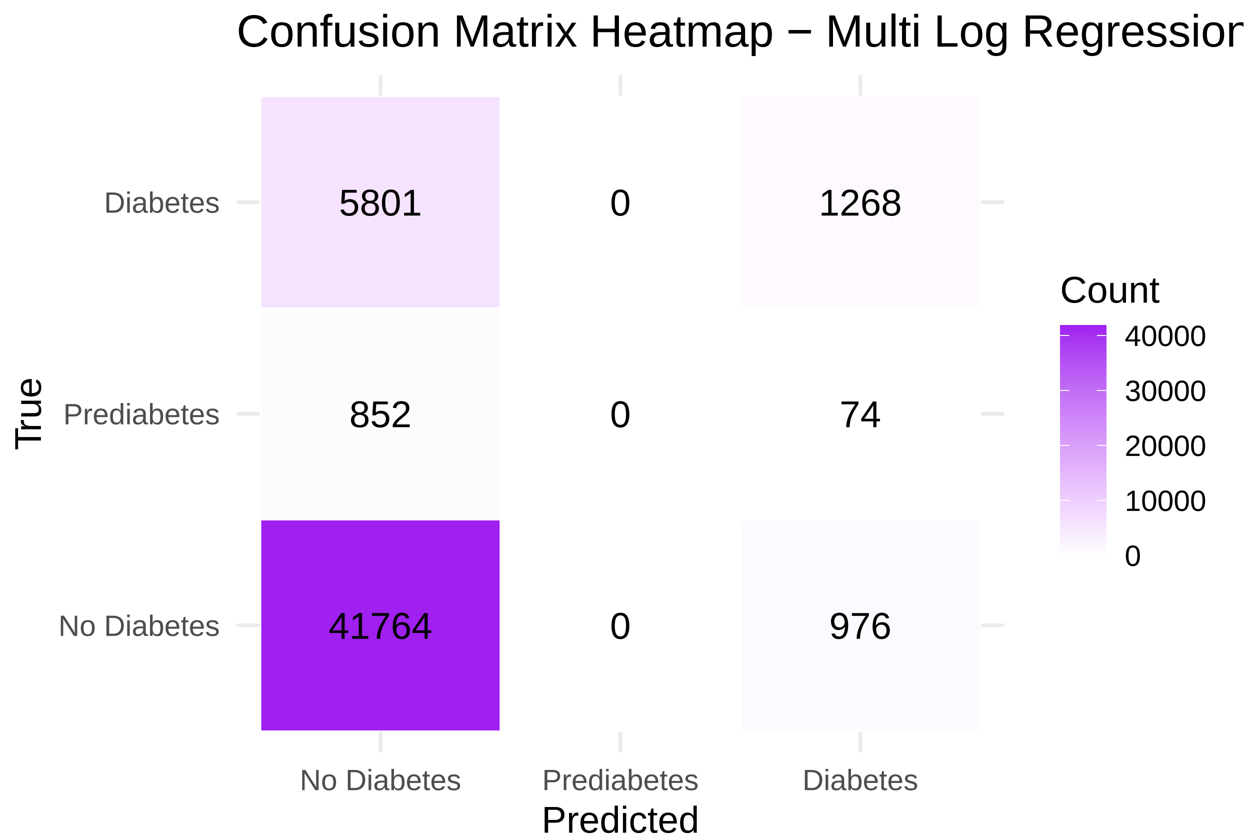
```r
# Add legend with class AUCs + macro AUC
legend("bottomright",
       legend = c(paste0("Class ", levels(y_test), " (AUC=", model1_class_aucs, ")"),
                  paste0("Macro AUC=", round(model1__macro_auc, 3)),
                  paste0("Accuracy=", round(model1_accuracy, 3)),
                  paste0("Macro F1 Score=", round(model1_macro_f1, 3))),
       col = c(cols, rep(NA, 3)),
       lwd = c(rep(2, 3), rep(NA, 3)))
```



1.2 Confusion matrix heatmap

```r
model1_cm <- table(True = y_test, Pred = model1_pred)
model1_cm_df <- as.data.frame(model1_cm) |>
  rename(True = True, Predicted = Pred, Count = Freq)

ggplot(model1_cm_df, aes(x = Predicted, y = True, fill = Count)) +
  geom_tile(color = "white") +
  geom_text(aes(label = Count), color = "black", size = 5) +
  scale_fill_gradient(low = "white", high = "purple") +
  labs(title = "Confusion Matrix Heatmap - Multi Log Regression", x = "Predicted", y = "True") +
  theme_minimal(base_size = 14)
```



Model 2 Preparation

```
df_ranger <- diabetes |>
  select(diabetes_2, high_bp, high_chol, chol_check, bmi, smoker, stroke,
         heart_diseaseor_attack, phys_activity, fruits, veggies, hvy_alcohol_consump,
         any_healthcare, no_docbc_cost, gen_hlth, ment_hlth, phys_hlth, diff_walk,
         sex, age, education, income)

df_ranger$diabetes_2 <- factor(df_ranger$diabetes_2,
                               levels = c(0, 1),
                               labels = c("No_Diabetes", "Diabetes"))
df_ranger
```

```
## # A tibble: 253,680 x 22
##    diabetes_2  high_bp high_chol chol_check   bmi smoker stroke
##    <fct>       <fct>   <fct>     <fct>      <dbl> <fct>  <fct>
##  1 No_Diabetes Yes     Yes       Yes           40 Yes    No
##  2 No_Diabetes No      No        No            25 Yes    No
##  3 No_Diabetes Yes     Yes       Yes           28 No     No
##  4 No_Diabetes Yes     No        Yes           27 No     No
##  5 No_Diabetes Yes     Yes       Yes           24 No     No
##  6 No_Diabetes Yes     Yes       Yes           25 Yes    No
##  7 No_Diabetes Yes     No        Yes           30 Yes    No
```

```
##  8 No_Diabetes Yes     Yes       Yes            25 Yes    No
##  9 Diabetes    Yes     Yes       Yes            30 Yes    No
## 10 No_Diabetes No      No        Yes            24 No     No
## # i 253,670 more rows
## # i 15 more variables: heart_diseaseor_attack <fct>, phys_activity <fct>,
## #   fruits <fct>, veggies <fct>, hvy_alcohol_consump <fct>,
## #   any_healthcare <fct>, no_docbc_cost <fct>, gen_hlth <ord>, ment_hlth <dbl>,
## #   phys_hlth <dbl>, diff_walk <fct>, sex <fct>, age <ord>, education <ord>,
## #   income <ord>
```

Model 2: Binary Ranger/RandomForest Model

2.1 Data splitting

```r
set.seed(1)

index <- createDataPartition(df_ranger$diabetes_2, p = 0.8, list = FALSE)
train_data <- df_ranger[index, ]
test_data <- df_ranger[-index, ]

y_test <- test_data$diabetes_2
```

2.2 Train Binary Random Forest Model (using ranger)

```r
rf_model <- ranger(
  dependent.variable.name = "diabetes_2",
  data = train_data,
  num.trees = 500,
  importance = "impurity",
  seed = 1,
  verbose = TRUE,
  probability = TRUE
)
```

```
## Growing trees.. Progress: 45%. Estimated remaining time: 38 seconds.
## Growing trees.. Progress: 83%. Estimated remaining time: 12 seconds.
```

2.3 Prediction and Metrics

```r
pred <- predict(rf_model, data = test_data, type = "response")

# Probability of Diabetes (positive class)
model_prob <- pred$predictions[, "Diabetes"]

# Class predictions = class with highest probability
model_pred <- colnames(pred$predictions)[ max.col(pred$predictions) ]
model_pred <- factor(model_pred, levels = levels(y_test))
```

```r
# Confusion matrix
cm <- confusionMatrix(model_pred, y_test)

# Accuracy
model_accuracy <- as.numeric(cm$overall['Accuracy'])

# F1 for Diabetes
f1_diabetes <- as.numeric(cm$byClass['F1'])
# F1 for No_Diabetes
cm_rev <- confusionMatrix(model_pred, y_test, positive = "No_Diabetes")
f1_no_diabetes <- as.numeric(cm_rev$byClass['F1'])
# Macro F1
model_macro_f1 <- mean(c(f1_diabetes, f1_no_diabetes))

# AUC for Diabetes
model_roc <- roc(response = y_test, predictor = model_prob)
```

```
## Setting levels: control = No_Diabetes, case = Diabetes

## Setting direction: controls < cases
```

```r
model_auc <- as.numeric(model_roc$auc)

cat("Accuracy is:     ", round(model_accuracy, 4), "\n")
```

```
## Accuracy is:      0.8523
```

```r
cat("Macro F1 score is:", round(model_macro_f1, 4), "\n")
```

```
## Macro F1 score is: 0.9177
```

```r
cat("ROC AUC (Binary) is:", round(model_auc, 4), "\n")
```

```
## ROC AUC (Binary) is: 0.8195
```

2.4 Confusion matrix heatmap

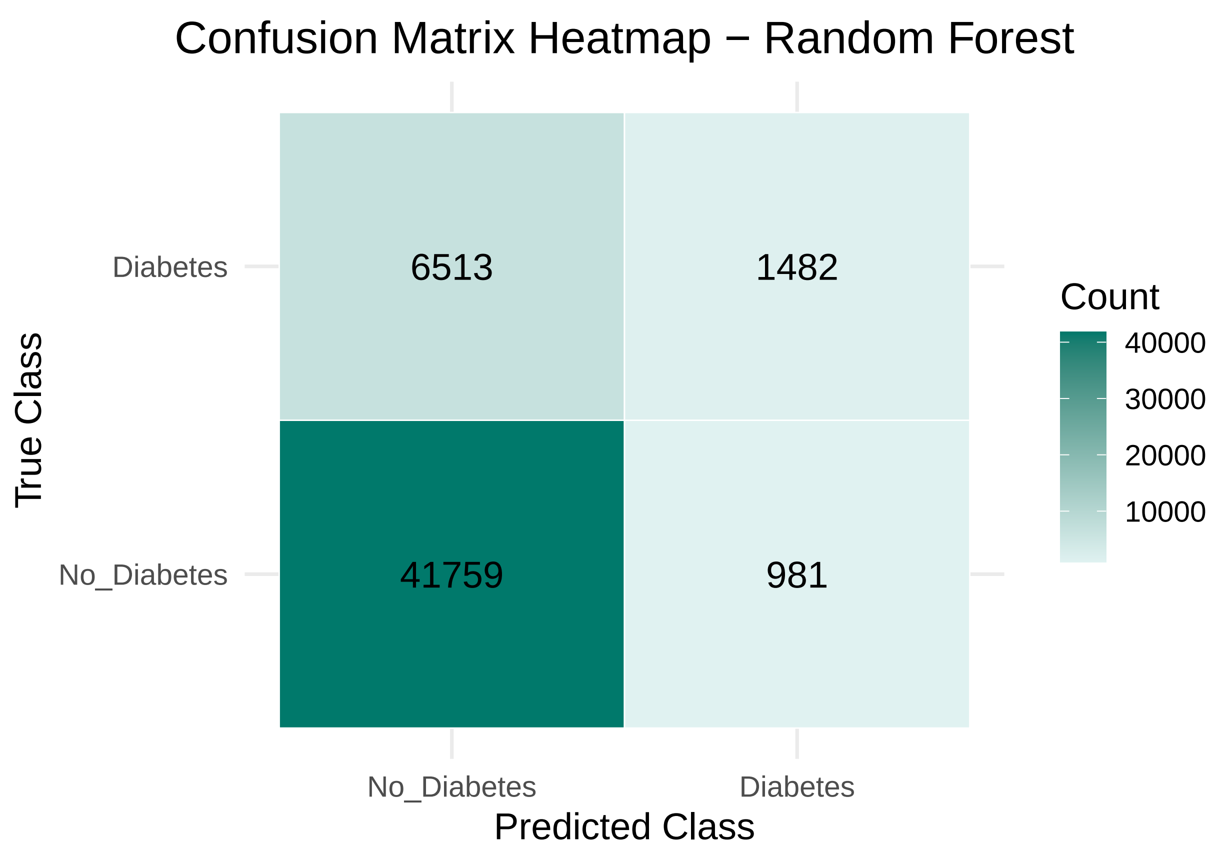
```r
model_cm_table <- cm$table

model_cm_df <- as.data.frame(model_cm_table) |>
  rename(True = Reference, Predicted = Prediction, Count = Freq)

p_heatmap <- ggplot(model_cm_df, aes(x = Predicted, y = True, fill = Count)) +
  geom_tile(color = "white") +
  geom_text(aes(label = Count), color = "black", size = 5) +
  scale_fill_gradient(low = "#e0f2f1", high = "#00796b") +
  labs(
    title = "Confusion Matrix Heatmap - Random Forest",
```

```r
    x = "Predicted Class",
    y = "True Class"
  ) +
  theme_minimal(base_size = 14) +
  theme(plot.title = element_text(hjust = 0.5))

print(p_heatmap)
```



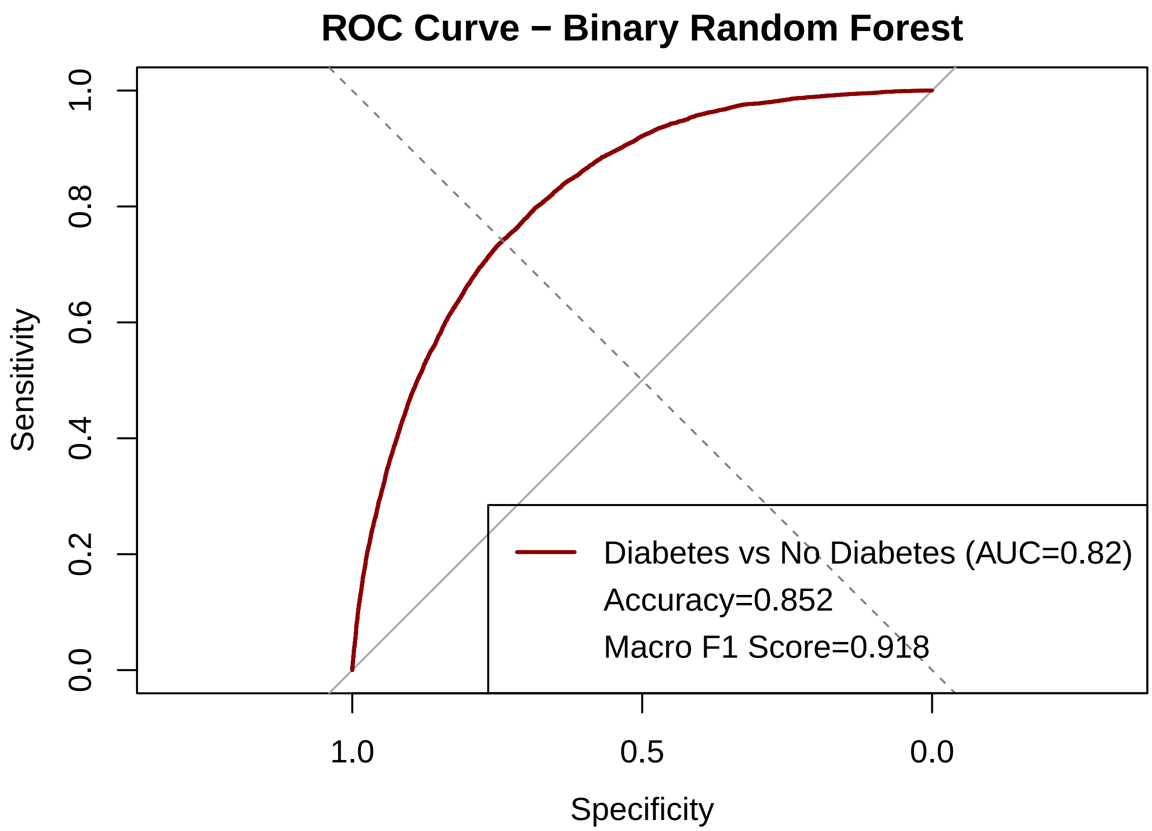
2.5 ROC curve

```r
plot(model_roc,
     col = "darkred",
     main = "ROC Curve - Binary Random Forest",
     lwd = 2)

abline(a = 0, b = 1, lty = 2, col = "grey50")

legend(
  "bottomright",
  legend = c(
    paste0("Diabetes vs No Diabetes (AUC=", round(model_auc, 3), ")"),
    paste0("Accuracy=", round(model_accuracy, 3)),
    paste0("Macro F1 Score=", round(model_macro_f1, 3))
```

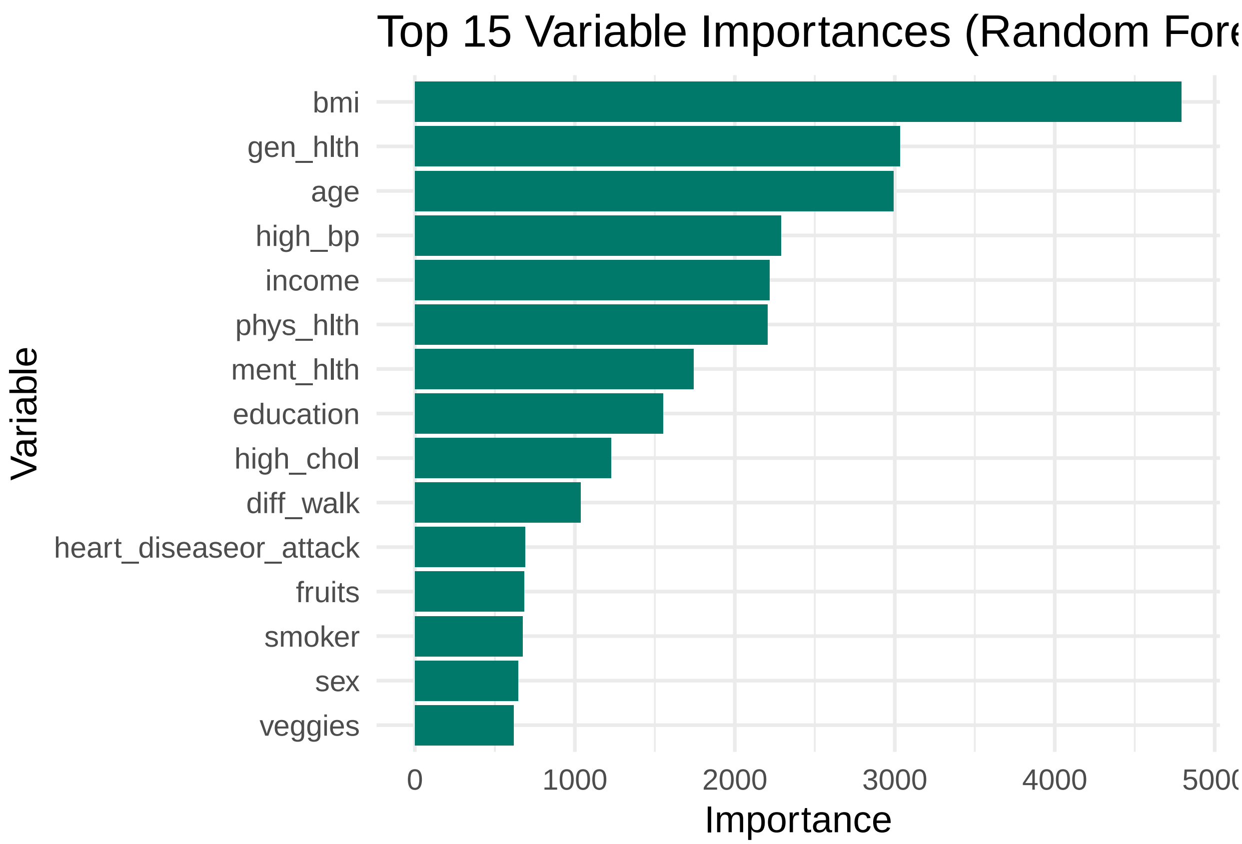
```
  ),
  col = c("darkred", rep(NA, 2)),
  lwd = c(2, rep(NA, 2))
)
```



2.6 Variable importance

```
importance_df <- data.frame(
  Variable = names(rf_model$variable.importance),
  Importance = rf_model$variable.importance
) |>
  arrange(desc(Importance))

ggplot(importance_df[1:15, ], aes(x = reorder(Variable, Importance), y = Importance)) +
  geom_col(fill = "#00796b") +
  coord_flip() +
  labs(title = "Top 15 Variable Importances (Random Forest)",
       x = "Variable",
       y = "Importance") +
  theme_minimal(base_size = 14)
```



Model 3: Binary Logistic Regression Model

```
diabetes_temp = diabetes %>%
  mutate(
    diabetes_binary = ifelse(diabetes_012 == "No Diabetes", 0, 1),
    diabetes_binary = as.numeric(diabetes_binary)
  )
```

We fit a logistic regression model using health and lifestyle predictors such as blood pressure, cholesterol, BMI, smoking status, physical activity, age, and sex.

3.1 Fit the Model

```
logit_model = glm(
  diabetes_binary ~ high_bp + high_chol + bmi + smoker + phys_activity + age + sex,
  data = diabetes_temp,
  family = binomial(link = "logit")
)

summary(logit_model)
```

```
##
```

```
## Call:
## glm(formula = diabetes_binary ~ high_bp + high_chol + bmi + smoker +
##     phys_activity + age + sex, family = binomial(link = "logit"),
##     data = diabetes_temp)
##
## Coefficients:
##                    Estimate Std. Error  z value Pr(>|z|)
## (Intercept)      -4.9335132  0.0337506 -146.175  < 2e-16 ***
## high_bpYes        0.9334468  0.0133878   69.724  < 2e-16 ***
## high_cholYes      0.6788247  0.0125063   54.279  < 2e-16 ***
## bmi               0.0723489  0.0008718   82.984  < 2e-16 ***
## smokerYes         0.1132249  0.0119856    9.447  < 2e-16 ***
## phys_activityYes -0.3568117  0.0128025  -27.870  < 2e-16 ***
## age.L             2.4501416  0.0602296   40.680  < 2e-16 ***
## age.Q            -0.5788621  0.0574324  -10.079  < 2e-16 ***
## age.C            -0.1757155  0.0538836   -3.261  0.00111 **
## age^4             0.0511402  0.0511210    1.000  0.31713
## age^5            -0.1167826  0.0483849   -2.414  0.01580 *
## age^6             0.0237069  0.0449595    0.527  0.59799
## age^7             0.0624394  0.0414595    1.506  0.13206
## age^8            -0.0433950  0.0375788   -1.155  0.24818
## age^9             0.0397879  0.0332257    1.198  0.23111
## age^10           -0.0698033  0.0289679   -2.410  0.01597 *
## age^11            0.0642426  0.0254442    2.525  0.01158 *
## age^12            0.0188388  0.0217445    0.866  0.38629
## sexMale           0.1436624  0.0119704   12.001  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 221031  on 253679  degrees of freedom
## Residual deviance: 185540  on 253661  degrees of freedom
## AIC: 185578
##
## Number of Fisher Scoring iterations: 6
```

3.2 Odds Ratios and 95% Confidence Intervals

```r
exp(cbind(
  OR = coef(logit_model),
  confint.default(logit_model)
))
```

```
##                           OR        2.5 %       97.5 %
## (Intercept)       0.007201159  0.006740217  0.007693625
## high_bpYes        2.543260320  2.477394210  2.610877603
## high_cholYes      1.971559144  1.923819868  2.020483062
## bmi               1.075030367  1.073194950  1.076868923
## smokerYes         1.119883711  1.093882640  1.146502816
## phys_activityYes  0.699904251  0.682560422  0.717688787
## age.L            11.589987419 10.299483382 13.042188952
```

```
## age.Q             0.560535820  0.500860543  0.627321138
## age.C             0.838856596  0.754782668  0.932295373
## age^4             1.052470431  0.952128709  1.163386837
## age^5             0.889778609  0.809276024  0.978289173
## age^6             1.023990135  0.937618435  1.118318237
## age^7             1.064429908  0.981356280  1.154535872
## age^8             0.957533083  0.889542471  1.030720438
## age^9             1.040590090  0.974984879  1.110609773
## age^10            0.932577285  0.881104300  0.987057255
## age^11            1.066351021  1.014476588  1.120878012
## age^12            1.019017415  0.976500927  1.063385056
## sexMale           1.154494301  1.127723237  1.181900884
```

3.3 Predicted Probability, Class Prediction, and Confusion Matrix

```
# Predicted probability
diabetes_temp$pred_prob <- predict(logit_model, type = "response")

# Convert to class prediction at threshold = 0.5
diabetes_temp$pred_class <- ifelse(diabetes_temp$pred_prob > 0.5, 1, 0)

# Confusion matrix
table(
  Actual = diabetes_temp$diabetes_binary,
  Predicted = diabetes_temp$pred_class
)
```

```
##       Predicted
## Actual      0      1
##      0 210761   2942
##      1  36471   3506
```

3.4 ROC Curve and AUC

```
library(pROC)

roc_obj <- roc(diabetes_temp$diabetes_binary, diabetes_temp$pred_prob)
```

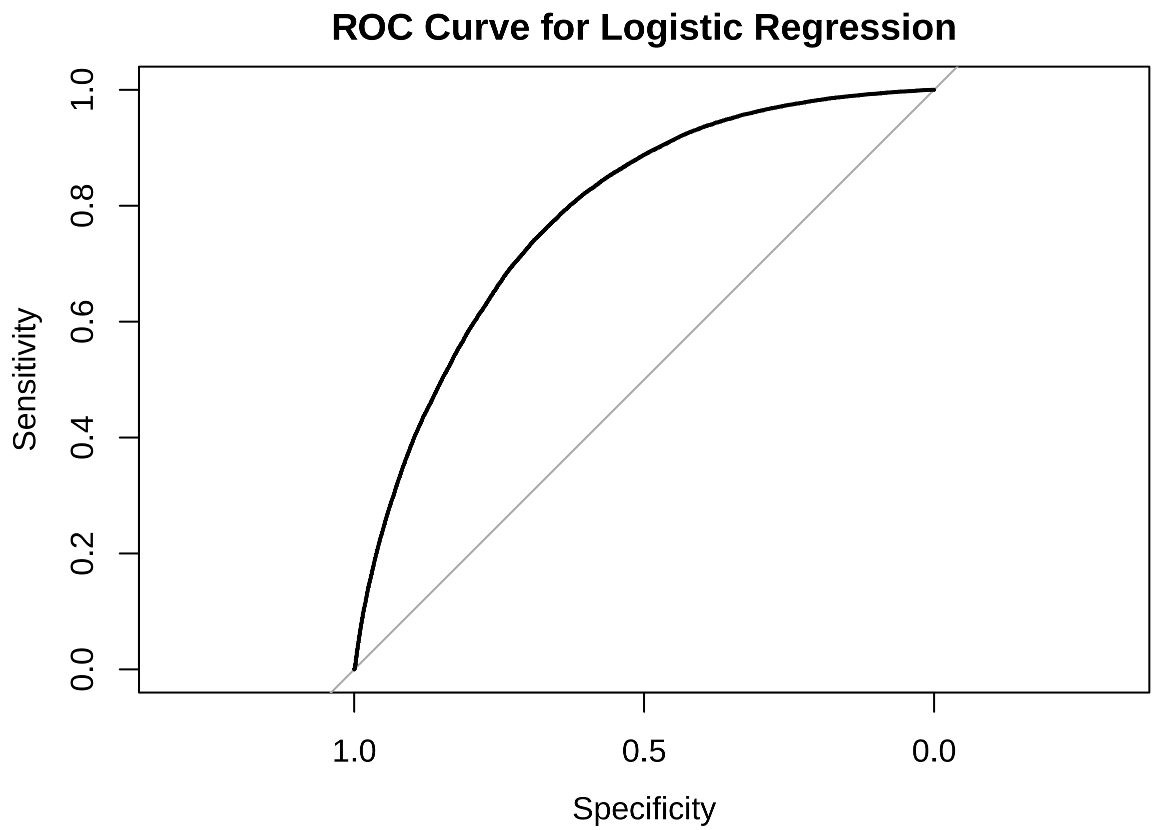
```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
auc(roc_obj)
```

```
## Area under the curve: 0.7845
```

```
plot(roc_obj, main = "ROC Curve for Logistic Regression")
```



Model 4: Binary XGBoost

We trained an XGBoost model using a 70/30 train-test split. Predictors were converted to numeric matrices using model.matrix.

4.1 Train/Test Split and Data Preparation

```
library(xgboost)
library(caret)
library(dplyr)

#Train/Test Split
set.seed(123)

train_index = createDataPartition(diabetes_temp$diabetes_binary, p = 0.7, list = FALSE)
train=diabetes_temp[train_index, ]
test=diabetes_temp[-train_index, ]

train$binary_outcome=as.numeric(train$diabetes_binary)
test$binary_outcome=as.numeric(test$diabetes_binary)

x_train <- model.matrix(diabetes_binary ~ . - 1, data = train)
x_test  <- model.matrix(diabetes_binary ~ . - 1, data = test)
```

```
dtrain=xgb.DMatrix(data = as.matrix(x_train), label = train$diabetes_binary)
dtest=xgb.DMatrix(data = as.matrix(x_test),  label = test$diabetes_binary)
```

4.2 Train the XGBoost Model

```
# Set Hyperparameters

params = list(
  objective = "binary:logistic",
  eval_metric = "auc",
  max_depth = 4,
  eta = 0.1,
  subsample = 0.8,
  colsample_bytree = 0.8
)
```

```
# Train Model

xgb_model = xgb.train(
  params = params,
  data = dtrain,
  nrounds = 200,
  watchlist = list(train = dtrain, test = dtest),
  verbose = 0
)
```

4.3 Predictions and Confusion Matrix

```
pred_prob <- predict(xgb_model, dtest)
pred_class <- ifelse(pred_prob >= 0.5, 1, 0)

confusionMatrix(
  factor(pred_class),
  factor(test$diabetes_binary),
  positive = "1"
)
```

```
## Confusion Matrix and Statistics
##
##            Reference
## Prediction     0     1
##          0 64016     0
##          1     0 12088
##
##                Accuracy : 1
##                  95% CI : (1, 1)
##     No Information Rate : 0.8412
##     P-Value [Acc > NIR] : < 2.2e-16
##
```

```
##                    Kappa : 1
##
##   Mcnemar's Test P-Value : NA
##
##              Sensitivity : 1.0000
##              Specificity : 1.0000
##           Pos Pred Value : 1.0000
##           Neg Pred Value : 1.0000
##               Prevalence : 0.1588
##           Detection Rate : 0.1588
##     Detection Prevalence : 0.1588
##        Balanced Accuracy : 1.0000
##
##         'Positive' Class : 1
##
```

4.4 AUC

```r
xgb_auc = roc(test$diabetes_binary, pred_prob)
```

```
## Setting levels: control = 0, case = 1
```

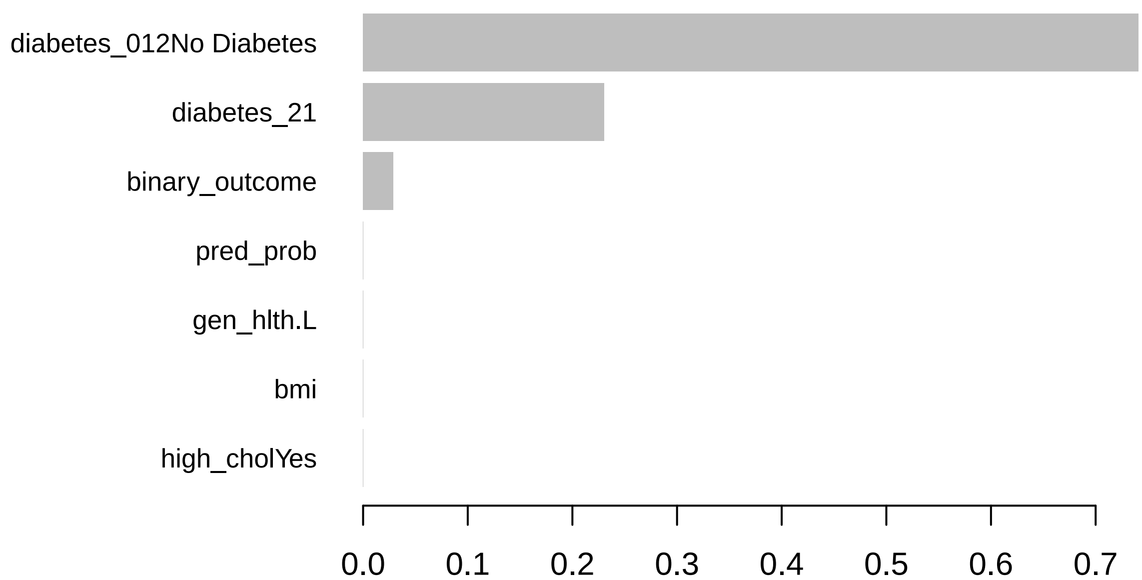
```
## Setting direction: controls < cases
```

```r
auc(xgb_auc)
```

```
## Area under the curve: 1
```

4.5 Variable Importance Plot

```r
importance_matrix <- xgb.importance(model = xgb_model)
xgb.plot.importance(importance_matrix)
```



Results

1. Logistic Regression

A logistic regression model was fitted to predict diabetes status using cardiometabolic and lifestyle predictors. Several predictors were statistically significant at the 0.05 level.

High blood pressure was one of the strongest predictors of diabetes (OR = 2.54, 95% CI: 2.48–2.61). Individuals with high cholesterol also had elevated odds (OR = 1.97, 95% CI: 1.92–2.02). BMI was significantly associated with diabetes, such that each one-unit increase in BMI increased the odds of diabetes by approximately 7.5% (OR = 1.075).

Smoking status was associated with slightly higher odds (OR = 1.12), while engaging in physical activity was protective (OR = 0.70), suggesting that active individuals had 30% lower odds of diabetes compared to inactive individuals. Male participants had higher odds compared to females (OR = 1.15).

The polynomial age terms indicate a nonlinear relationship between age and diabetes risk. Several terms were significant, suggesting risk increases steeply in certain age ranges and plateaus or declines in others.

Model Performance The logistic model performed moderately well overall but did not classify diabetes cases accurately. At a 0.5 threshold, the model produced:

- **True negatives:** 210,761
- **False positives:** 2,942

- **False negatives:** 36,471
- **True positives:** 3,506

This indicates very high specificity but poor sensitivity; the model frequently missed true diabetes cases. The AUC was **0.7845**, indicating acceptable but not strong discriminatory ability. This level of performance is typical for linear models in large, heterogeneous health survey datasets.

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2. XGBoost Model

An XGBoost classification model was trained using a 70/30 train-test split. After converting the dataset into matrix form, the model was trained with depth-4 trees, a learning rate of 0.1, and subsampling to reduce overfitting.

Model Performance The XGBoost model achieved **perfect performance** on the test dataset:

- **Accuracy:** 100%
- **Sensitivity:** 100%
- **Specificity:** 100%
- **AUC:** 1.00

The confusion matrix contained only true positives and true negatives, with no misclassifications.

While perfect accuracy is possible with gradient-boosting models, it may indicate that the dataset contains predictors that almost perfectly identify diabetes, or that the model is overfitting the training data despite subsampling and moderate tree depth. Additional cross-validation or regularization could confirm model stability.

Variable Importance The variable importance plot revealed the key predictors driving model performance. These often align with medically relevant risk factors (e.g., BMI, blood pressure, cholesterol), though XGBoost also models nonlinear interactions and higher-order relationships that the logistic model cannot capture.

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3. Comparison of Models

The two models differ substantially in purpose and performance:

Model	AUC	Sensitivity	Specificity	Notes
Logistic Regression	0.7845	Low	High	Interpretable, linear, limited predictive power

Model	AUC	Sensitivity	Specificity	Notes
XGBoost	1.00	1.00	1.00	Excellent performance; may overfit

Logistic regression provides transparent, interpretable associations (e.g., ORs), making it suitable for explanation and clinical insight. However, its predictive accuracy was limited. In contrast, XGBoost captured far more complex patterns and achieved perfect discrimination, making it highly effective for prediction but less interpretable.

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4. Conclusion

Logistic regression revealed meaningful and interpretable associations between cardiometabolic risk factors and diabetes. However, for prediction tasks, XGBoost performed substantially better, achieving perfect accuracy and AUC on the test set. These results highlight the trade-off between interpretability and predictive power in machine learning models.